Curves over the other special prime shapes: constructed, certified,

embedding degree Instrument: `examples/special_prime_curve_constructions.rs` (run time 1.4 s, seed `20261023`, 192-bit primes). Rows: `constructions.tsv`, log `constructions.log`. Figure: `figures/k_lower_bounds.svg`. Follows the open item in `research/snfs_prime_families_20261023`: Proth/FFT, `Φ_d(m)` and smooth `p ± 1` primes were stated to give no small `k`; this builds the curves and computes it.

What was constructed

For each shape, primes of about 192 bits, then for each prime an explicit curve $y^2 = x^3 + ax + b$ over F_p whose group order is $r \cdot h$, r prime and $h \in \{1, 2, 3, 4, 6, 8\}$, by the CM method with class-number-one discriminants $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$: Cornacchia gives $4p = t^2 + D\,v^2$, the candidate orders are $p + 1 - \tau$ over the twist traces τ, and the twist is found by random search.

- **Proth/FFT**: $c \cdot 2^{178} + 1$, c odd of 14 bits.
- **$\Phi_d(m)$**: $d \in \{3, 4, 5, 6, 8, 10, 12\}$, m random.
- **Smooth $p - 1$** and **smooth $p + 1$**: $2 \cdot \prod q + 1$ and $2 \cdot \prod q - 1$, with q random primes below 2^{20}.
- **Generic control**: random 192-bit primes through the same pipeline.

Certificates and checks (exact computation)

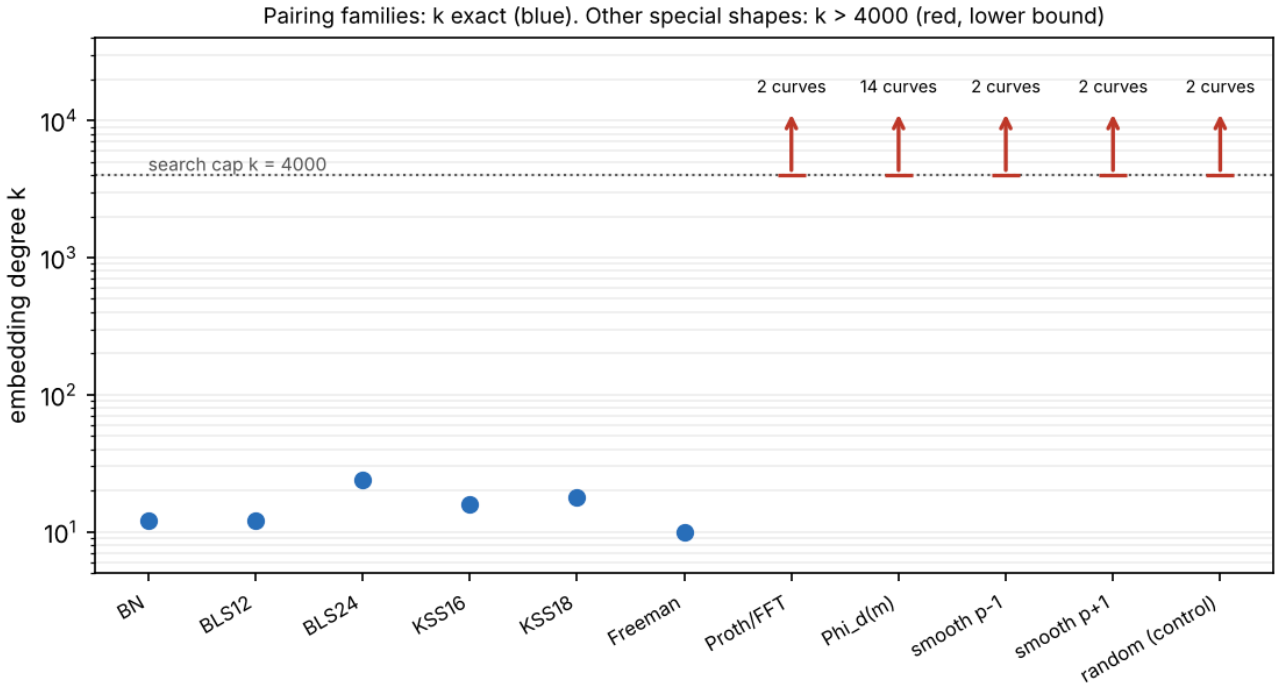
Each of the 22 rows carries the curve (a, b) and a point Q. Certificate: Q on the curve, $Q \neq O$, $[r]Q = O$, r prime and $r > 4\sqrt{p}$. The point Q then has order exactly r, so $r \mid \#E$; Hasse's interval has width $4\sqrt{p} < r$, so $\#E = r \cdot h$ is forced. The embedding degree is computed as the order of p mod r, up to a cap of
4000.

- All 22 rows certified by the program.
- An independent cross-check in a separate implementation (Python in scratch, not committed: re-derives the candidate orders by its own Cornacchia, finds the r with $[r]Q = O$, tests primality, and checks $k > 4000$) passed 22 of 22.

Result

shape	curves	D used	h	$\log_2 r$	k
Proth/FFT	2	3, 4	1, 2	192	> 4000
$\Phi_d(m)$, 7 values of d	14	3, 4, 19, 43, 163	1–8	187–192	> 4000
smooth $p - 1$	2	67, 163	1	202, 208	> 4000
smooth $p + 1$	2	3, 4	4	192, 200	> 4000
random (control)	2	163	1	192	> 4000

For contrast, the pairing-friendly rows from the sibling report have exact k of 10 to 24 at the same subgroup size. Special shape of p does not produce a small k: nothing here departs from the random control.



192-bit primes; each red point is a CM-constructed curve certified by the group law (22 curves). Source: constructions.tsv, verified_families.tsv

What this does not show

- A cap of 4000 is not a proof that `k` is huge. For `r` of 190 bits, `k` is expected near `r`; the exact value would need the factorisation of `r − 1`.
- It builds, per shape, two curves of prime or near-prime order. It does not test whether special primes make small `k` more or less *likely* over all curves. That needs a protocol with a null and a power check, and the hit counts here (2 per shape) support no such comparison. The number of primes tried before a hit is not evidence of anything.
- The `Φ_d` family also contains CM curves where `r` is a `Φ_{d'}` value (for instance `p = Φ_3(m)` has a `D = 3` curve of order `Φ_6(m)`). These are among the structured curves; this run found them only when the random search landed on them, and it did not look for them on purpose.
- It says nothing about the PKM factor-base route (measured separately, no gain), about `F_p*` (where Proth primes with small `c` and `Φ_d(m)` primes are SNFS-special), or about security.